

FOSSEE DWSIM Internship Screening

Stage 1 – Screening Task Submission

SCREENING TASK 3

Physics-Constrained Surrogate Modeling of a Binary Distillation Column

using DWSIM and Machine Learning

System	Benzene–Toluene binary distillation column
Thermodynamics	Peng–Robinson (PR), DWSIM 9.0.5
Dataset	10,000 converged simulations + 340 failed attempts
Selected surrogate	128–64 ReLU ANN with material-balance projection

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Section 1

Problem Statement and Engineering Scope

The objective is to replace repeated steady-state DWSIM calculations with a fast surrogate for a benzene–toluene distillation column. The model predicts two product purities and two energy duties from operating and design variables. It is deliberately treated as an *emulator of the submitted DWSIM configuration*; it is not presented as a plant model and has not been validated against experimental data.

Inputs used by the surrogate

- Feed temperature, K
- Feed pressure, Pa
- Feed benzene mole fraction
- Number of column stages
- Relative feed-stage location
- Reflux ratio
- Bottoms withdrawal rate, mol/s

Predicted outputs

- Distillate benzene mole fraction
- Bottoms toluene mole fraction
- Condenser duty, kW
- Signed reboiler duty, kW

The final workflow emphasizes three questions beyond raw accuracy: whether the model generalizes to unseen data, whether its purity predictions respect the component material balance, and whether local trends agree with fresh DWSIM calculations.

Section 2

DWSIM Model

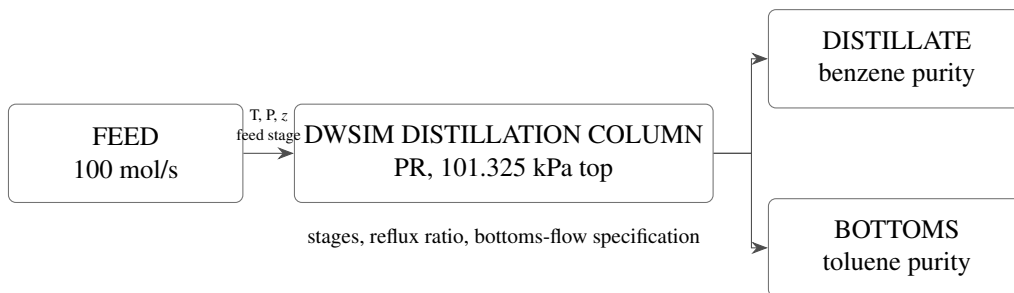


Table 1: Submitted DWSIM flowsheet configuration.

Item	Configuration
Components	Benzene and toluene
Thermodynamic model	Peng–Robinson (PR)
DWSIM version	9.0.5
Feed flow	100 mol/s, fixed
Column pressure	101,325 Pa top pressure; zero specified pressure drop
Condenser specification	Reflux ratio
Reboiler specification	Bottoms product molar flow, mol/s
Stage convention	DWSIM counts the condenser as stage index 0
Solver limit	300 column iterations
Duty convention	Signed DWSIM automation values in kW

The signed duty convention is retained. Condenser duty is positive throughout the successful dataset; reboiler duty is negative in most cases but positive in 94 simulations. Taking an absolute value would therefore remove information around a change in the simulated energy balance.

Section 3

Dataset Generation and Audit

The dataset is generated through `Code/generate_dataset.py`. Each case creates and solves a DWSIM flowsheet through the Automation API. Seven variables are sampled using independent 300-point Latin hypercube batches. Failed solves are written to `failed_cases.csv` and replaced by later samples until 10,000 converged rows are collected.

Table 2: Operating domain used for data generation.

Input	Range	Unit
Feed temperature	345–385	K
Feed pressure	101,325–200,000	Pa
Benzene feed mole fraction	0.25–0.75	–
Number of stages	18–35	–
Sampled relative feed location	0.30–0.70	–
Realized relative feed location	0.27778–0.72222	–
Reflux ratio	1.5–4.5	–
Bottoms flow	25–75	mol/s

Dataset audit	
Converged simulation rows	10,000
Failed DWSIM attempts retained for analysis	340
Overall failure fraction	3.29%
Missing or non-finite model values	0
Duplicate model-input rows	0
Distillate purities ≥ 0.999	35.61%
Bottoms purities ≥ 0.999	39.31%

The failed-case analysis shows that the nominal hypercube is not uniformly easy to solve. The highest observed quintile failure rates are 10.59% at low bottoms flow and 10.46% at high stage count. The regression domain is therefore best understood as the *converged portion* of the nominal operating space.

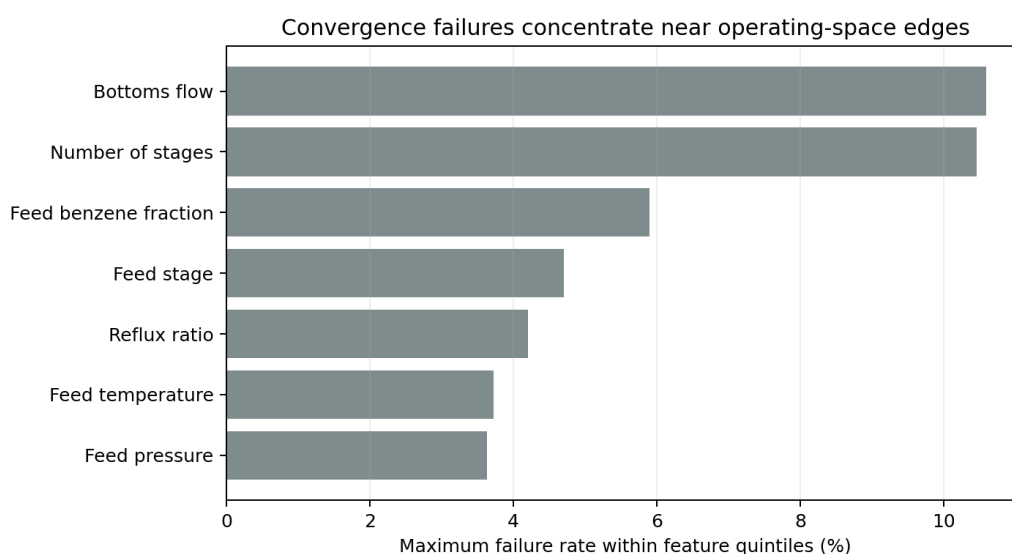


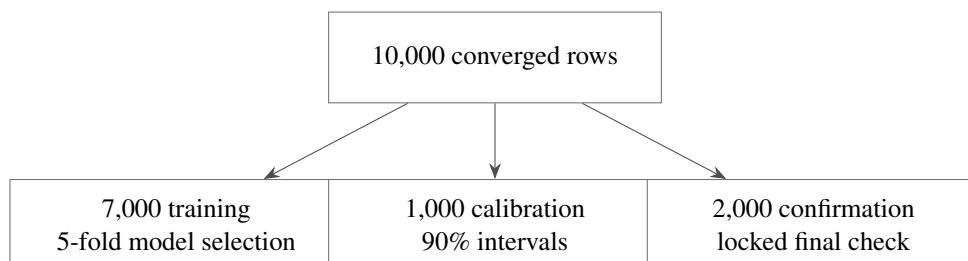
Figure 1: Maximum observed failure rate within quintile bins for each sampled input.

Section 4

Preprocessing and Validation Design

The raw feed-stage index and normalized feed-stage location are deterministic functions of one another when stage count is known. To avoid duplicate representation, the final surrogate uses stage count and the normalized feed-stage fraction, but not the raw feed-stage index. Standardization is fitted inside the scikit-learn estimator pipelines so preprocessing is learned only from the relevant training fold.

The 10,000 successful rows are assigned once using random seed 20260817:



NRMSE is defined as RMSE divided by the corresponding target range in the 7,000-row training set. Model family and ANN size are selected only from five-fold cross-validation on these training rows. The 1,000 calibration rows are used only for split-conformal prediction intervals, while the 2,000 confirmation rows remain untouched until final scoring.

4.1 Candidate models

Table 3: Model families evaluated in `Code/main.py`.

Family	Implementation
Polynomial regression	Quadratic polynomial features + Ridge regression
Tree ensemble	Extra Trees, 500 estimators
Gradient boosting	CatBoost, 500 iterations, depth 7
Artificial neural network	64–32, 128–64, and 256–128 ReLU MLPs plus a 128–64 tanh MLP
Physics-latent ANN	Bounded latent benzene split + two duty outputs

Section 5

Physics-Constrained Purity Prediction

The two product purities are coupled by the benzene component balance. With feed flow $F = 100$ mol/s, bottoms flow B , distillate flow $D = F - B$, feed benzene fraction z , distillate benzene purity $x_{D,Bz}$, and bottoms toluene purity $x_{B,Tol}$,

$$Fz = Dx_{D,Bz} + B(1 - x_{B,Tol}). \quad (1)$$

An unconstrained regressor predicts both purities independently and therefore implies two estimates of benzene flow in the distillate:

$$m_1 = D\hat{x}_{D,Bz}, \quad m_2 = Fz - B(1 - \hat{x}_{B,Tol}). \quad (2)$$

The selected model applies the explicit least-change projection implemented in the submitted physics-surrogate module. The projected component flow is

$$m^* = \frac{m_1/D^2 + m_2/B^2}{1/D^2 + 1/B^2}, \quad (3)$$

then clipped only to the physically feasible component-flow interval $[\max(0, Fz - B), \min(Fz, D)]$. Both purities are recomputed from the projected flow. This is not hidden independent clipping of purity predictions; it enforces a single physically consistent component split.

A second, fully latent constrained ANN was also evaluated. It predicts a bounded benzene split directly through a logistic transform, but it ranked below the projected ANN in cross-validation.

Section 6

Model Comparison and Selection

Table 4: Five-fold development-set comparison. Lower mean NRMSE is better.

Rank	Candidate	Mean NRMSE	Fold SD	Mean R^2
1	ANN 128–64 ReLU	0.005410	0.000392	0.999432
2	ANN 256–128 ReLU	0.005652	0.000270	0.999376
3	ANN 64–32 ReLU	0.006305	0.000573	0.999235
4	ANN 128–64 tanh	0.006630	0.000081	0.999146
5	ANN latent split	0.010421	0.003173	0.997631
6	CatBoost	0.010601	0.000466	0.997077
7	Extra Trees	0.021993	0.001183	0.981505
8	Polynomial Ridge	0.029151	0.000682	0.980632

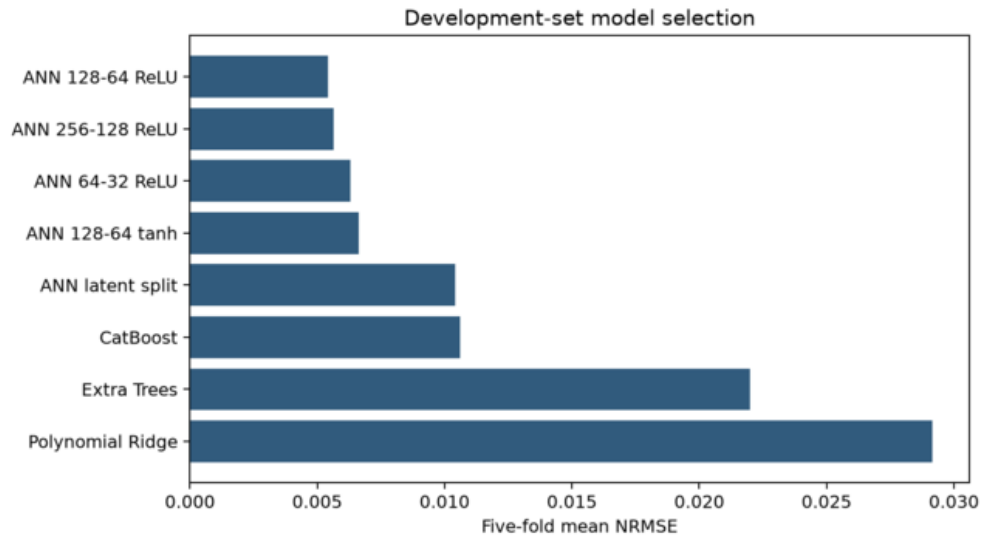


Figure 2: Five-fold model comparison used to select the final surrogate.

The 128–64 ReLU ANN is selected because it has the lowest development-set mean NRMSE. Selection is completed before any confirmation-set score is used.

Section 7

Locked Confirmation Performance

Table 5: Performance on the untouched 2,000-row confirmation set.

Output	MAE	RMSE	R^2	NRMSE	P95 abs. error
Distillate benzene purity	0.002004	0.005473	0.999053	0.008259	0.007373
Bottoms toluene purity	0.001653	0.003570	0.999633	0.005403	0.005685
Condenser duty (kW)	28.944	37.209	0.999755	0.003330	73.155
Signed reboiler duty (kW)	38.852	57.423	0.999529	0.003984	110.186

Headline confirmation result	
Mean NRMSE across four outputs	0.005244
Mean R^2 across four outputs	0.999493
Final invalid purity predictions	0 / 4,000
Maximum final benzene-balance residual	1.42×10^{-14} mol/s

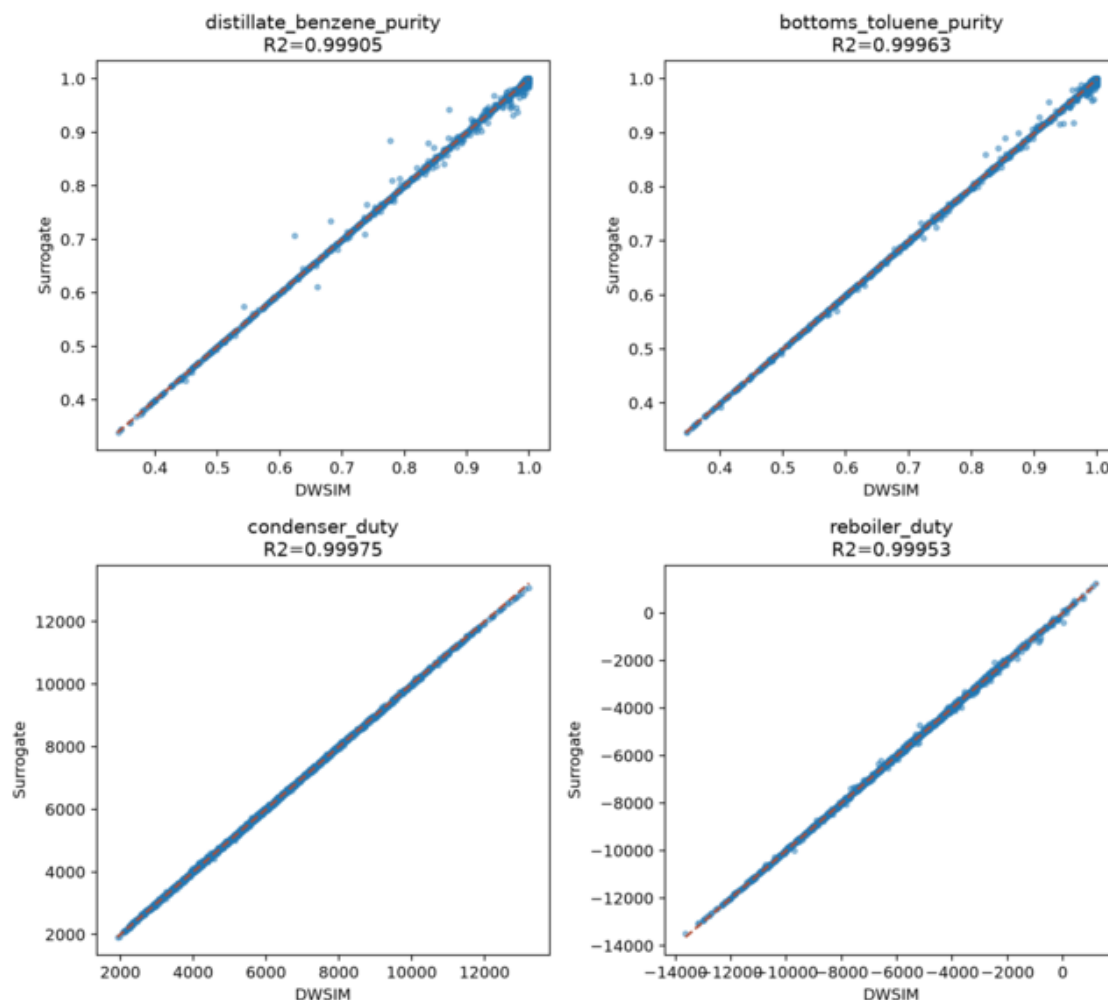


Figure 3: Locked confirmation predictions against DWSIM outputs.

7.1 Performance away from purity saturation

Global purity scores include many near-pure cases. Each purity was therefore rescored only where the corresponding DWSIM purity was below 0.995. The distillate subset contains 1,133 rows with RMSE 0.006943 and $R^2 = 0.998419$; the bottoms subset contains 1,084 rows with RMSE 0.004399 and $R^2 = 0.999380$.

7.2 Physical correction and uncertainty

Before projection, the same 128–64 ANN produced 257 invalid distillate purities and 250 invalid bottoms purities on the confirmation rows. The projected surrogate produces none. Mean absolute projection changes are 0.002250 for distillate purity and 0.001935 for bottoms purity; the corresponding 95th-percentile changes are 0.006614 and 0.005467.

Split-conformal intervals are calibrated using the separate 1,000-row calibration set:

Table 6: Empirical coverage of nominal 90% prediction intervals.

Output	Nominal	Observed	Mean interval width
Distillate purity	90%	91.4%	0.00815
Bottoms purity	90%	91.4%	0.00705
Condenser duty	90%	89.7%	121.85 kW
Reboiler duty	90%	89.0%	152.21 kW

Section 8

Robustness Beyond a Random Split

Random-split interpolation is intentionally supplemented by harder edge-region tests. For each test, the highest operating-region quintile is removed from fitting and used as an unseen test region.

Table 7: One-sided edge-region holdout tests.

Withheld region	Mean NRMSE	Important observation
Highest 20% feed benzene fraction	0.004973	Distillate-purity $R^2 = 0.98587$
Highest 20% reflux ratio	0.006226	Duty errors increase at the edge
Highest stage-count region	0.005190	2,505 rows held out because of integer ties
Highest 20% bottoms flow	0.011627	Distillate-purity $R^2 = 0.94375$

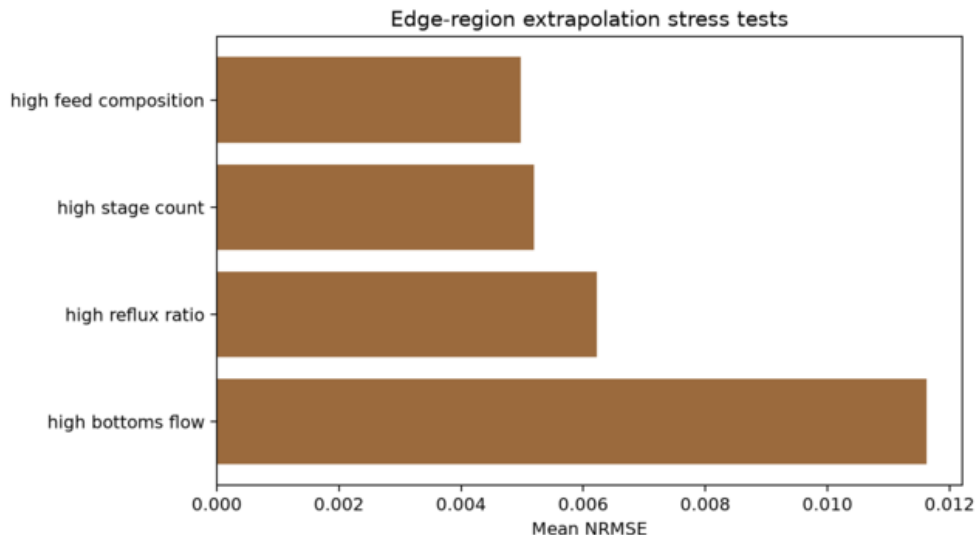


Figure 4: Edge-region stress tests reveal behavior hidden by random splitting.

8.1 Fresh DWSIM-versus-surrogate sweeps

The updated submission adds `Code/run_sweep_validation.py`, which solves fresh DWSIM points and compares them directly with the saved surrogate. A representative case is fixed at 365 K feed temperature, 150,662.5 Pa feed pressure, 0.5 benzene feed fraction, midpoint feed location, and 60 mol/s bottoms flow.

First, reflux ratio is swept through 13 values from 1.5 to 4.5 at 26 stages. The duty curves nearly overlap; the purity predictions follow the initial rise but show a small low bias and oscillation once the DWSIM purities approach their plateau.

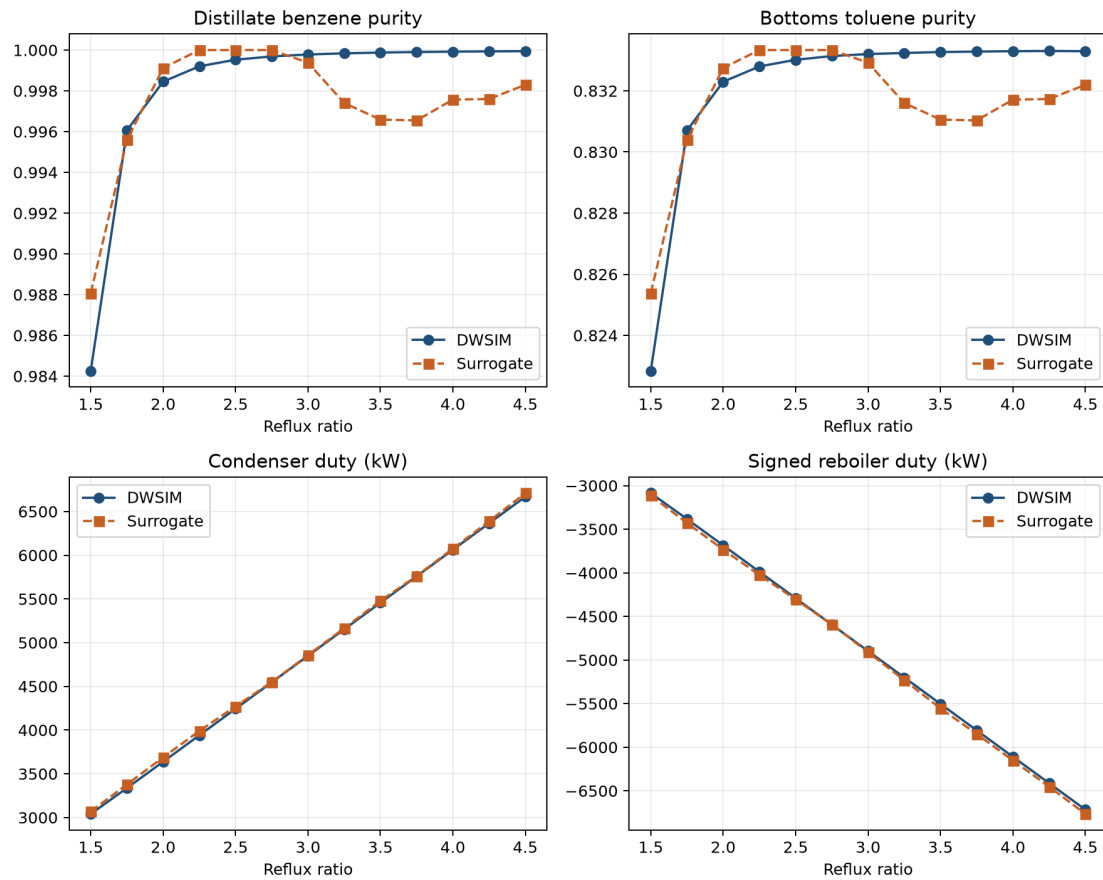


Figure 5: Fresh DWSIM and surrogate calculations for a reflux-ratio sweep.

Second, stage count is swept through every integer from 18 to 35 at reflux ratio 3.0, with feed stage moved to the nearest midpoint. The surrogate remains close in aggregate error but does not reproduce the exact DWSIM purity plateau, and small duty offsets remain visible.

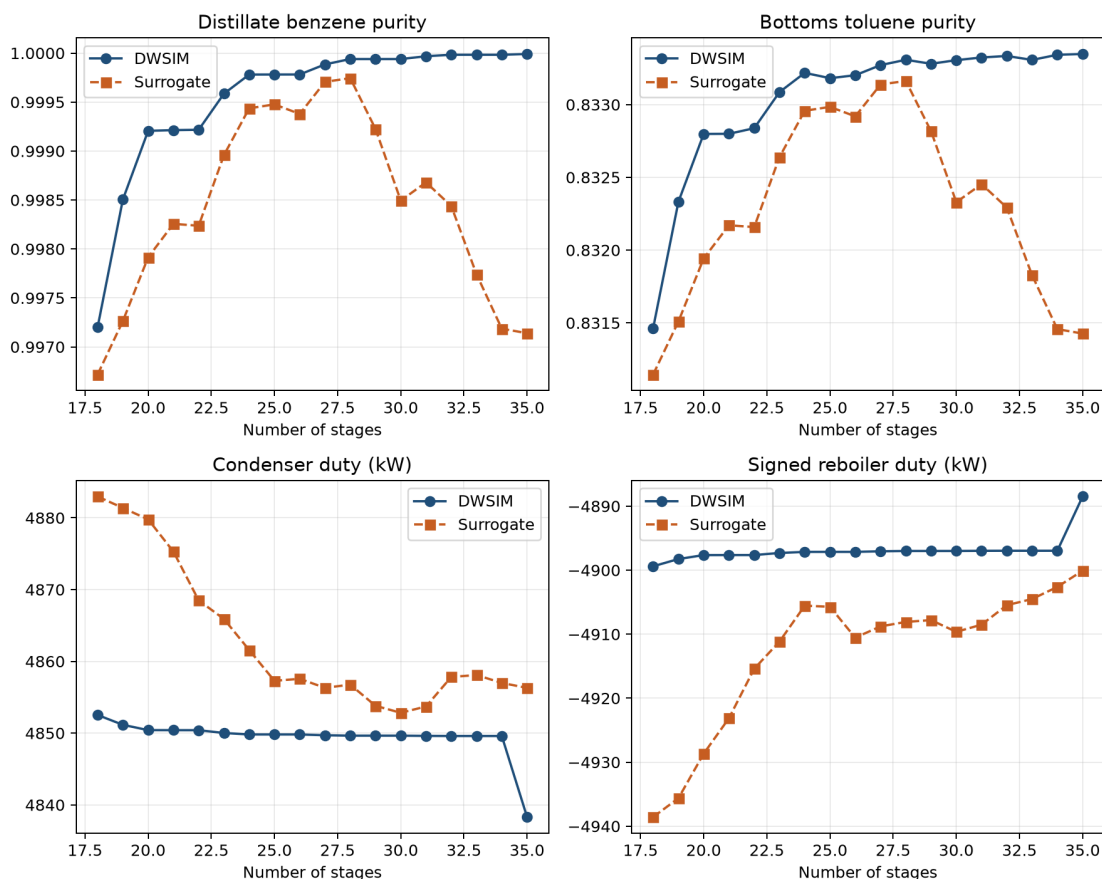


Figure 6: Fresh DWSIM and surrogate calculations for a stage-count sweep.

Across the two sweeps, all 31 reference points are solved again through DWSIM Automation rather than reused from the training CSV. This is a useful local validation because high global R^2 does not guarantee exact shape agreement near saturated purity regions.

8.2 Broader surrogate-only trend checks

One hundred confirmation conditions are also swept over reflux ratio and stage count while other model inputs remain fixed. Condenser duty increases and signed reboiler duty decreases with reflux in every tested step. Purity monotonicity is approximate: 0.83% of distillate reflux steps and 0.43% of bottoms reflux steps decrease by more than 0.001 purity. No stage-count step decreases either purity by more than 0.001. Balance and purity bounds are exact; monotonicity is therefore reported as a measured limitation rather than assumed.

Section 9

Reproducibility and Submission Structure

The final archive is organized around reproducible entry points rather than shipping every generated intermediate result. The detailed audit CSVs, plots, split assignments, and fitted model are regenerated by the submitted scripts.

Table 8: Principal files in the final submission.

File	Purpose
Code/generate_dataset.py	DWSIM Automation, Latin-hypercube sampling, convergence logging
Code/main.py	Model comparison, locked validation, uncertainty, robustness, final model fitting
Code/run_sweep_validation.py	Fresh DWSIM-versus-surrogate reflux and stage sweeps
Code/physics_surrogate.py	Benzene-balance projection and bounded latent alternative
Code/predict_surrogate.py	Guarded inference for new operating cases
Code/ml_model_comparison.ipynb	Short exploratory model-comparison walkthrough
dataset.csv	10,000 converged simulation rows
failed_cases.csv	340 failed attempts retained for convergence analysis
DWSIM_Flowsheet.dwxmz	Representative solved DWSIM flowsheet
Results_Summary.txt	Consolidated metrics and representative predictions

The main analysis is reproduced with:

```
python -m venv venv
venv/bin/python -m pip install -r Code/requirements.txt
venv/bin/python Code/main.py -results-dir reproduced_results
venv/bin/python Code/run_sweep_validation.py
    -model reproduced_results/final_surrogate.joblib
    -results-dir reproduced_results
```

Dataset regeneration additionally requires DWSIM on Linux. The README documents DWSIM discovery, dataset regeneration, training, sweep validation, and guarded prediction on new CSV inputs.

Section 10

Limitations and Conclusion

- The surrogate is valid only for the submitted benzene–toluene, Peng–Robinson, fixed-feed-flow and fixed-column-pressure setup.
- Regression results are conditional on DWSIM convergence; the surrogate does not predict numerical feasibility.
- Axis-aligned range checks do not detect every sparse corner inside the seven-dimensional box; split-conformal intervals also assume future interpolation cases are exchangeable with calibration data.
- The model has not been validated against experimental or plant data and must not be described as a plant model.
- Purity monotonicity is approximate near saturation, although large reversals are uncommon in the tested sweeps.

The **128–64 ReLU ANN with material-balance projection** is selected because it gives the lowest cross-validated normalized error while enforcing exact purity bounds and benzene balance. On 2,000 locked confirmation rows it achieves mean NRMSE 0.005244 and mean R^2 0.999493. Its credibility also comes from measured uncertainty, convergence conditioning, edge-region tests, and fresh DWSIM sweep comparisons.

Final engineering conclusion. The submitted surrogate is a high-accuracy interpolative emulator of the specified DWSIM column. Its scope, physical constraints, convergence domain, uncertainty, and extrapolation weaknesses are explicitly documented.